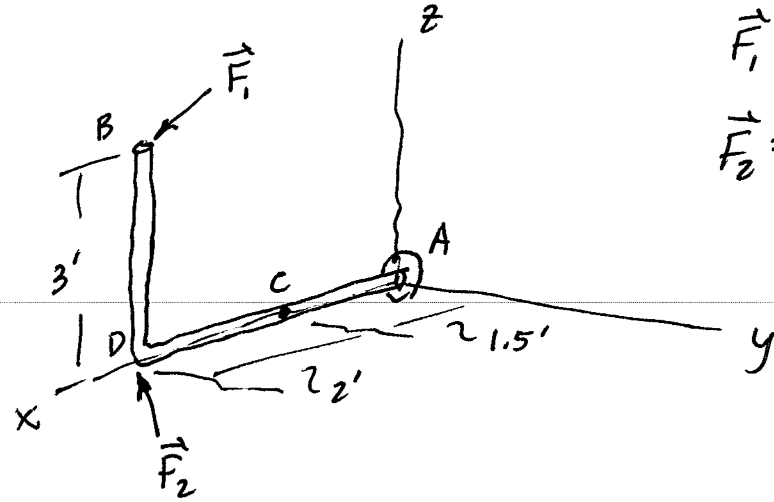
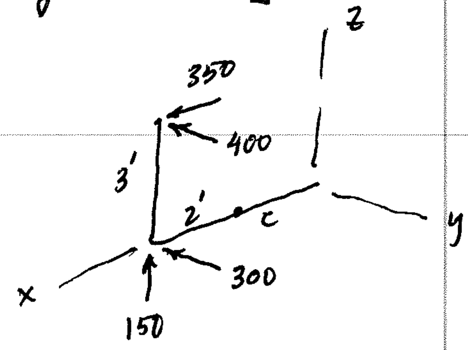
[Worked as a vector problem]

7-41 3-D Internal Forces

$\vec{F}_1 = [350\hat{i} - 400\hat{j}]$ lb

$\vec{F}_2 = [-300\hat{j} + 150\hat{k}]$ lb





$$\vec{F}_R \text{ at } C = \vec{F}_1 + \vec{F}_2$$

$$= [350\hat{i} - 700\hat{j} + 150\hat{k}] \text{ lb}$$

$$\vec{M}_{RC} = \Sigma \vec{r} \times \vec{F}$$

$$\vec{r}_{CD} = [2, 0, 0] \text{ ft}$$

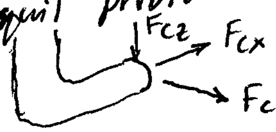
$$\vec{r}_{CB} = [2, 0, 3] \text{ ft}$$

$$\vec{M}_{RC} = [2,0,0] \times [0, -300, 150] + [2,0,3] \times [350, -400, 0]$$

$$\vec{M}_{RC} = [0, -300, -600] + [1200, 1050, -800]$$

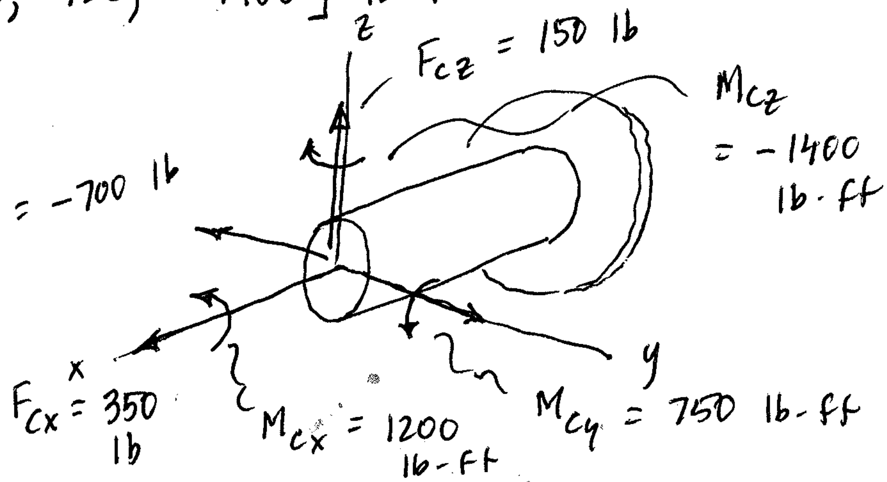
$$\vec{M}_{RC} = [1200, 750, -1400] \text{ lb-ft}$$

Book's signs are all neg of mine bc they worked it as an equil problem



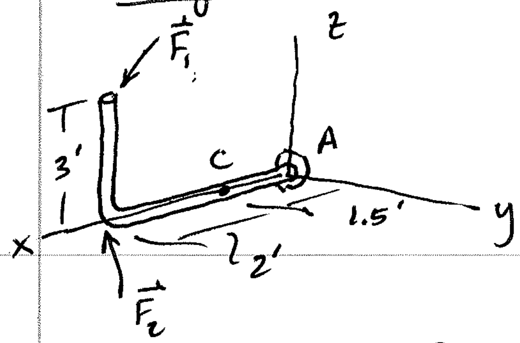
etc.

$F_{Cy} = -700$ lb



7-41 3 D Internal Forces Problem

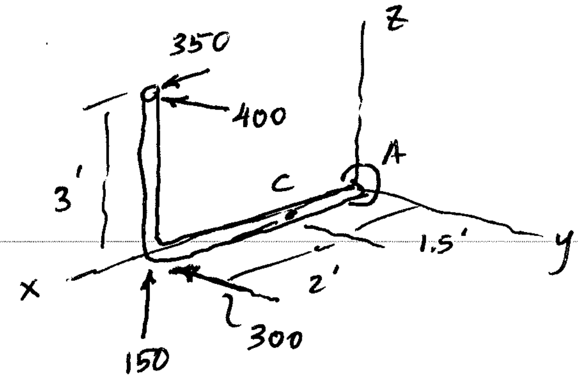
As given:



$\vec{F}_1 = [\,350\hat{i} - 400\hat{j}\,]$ lb

$\vec{F}_2 = [\,-300\hat{j} + 150\hat{k}\,]$ lb

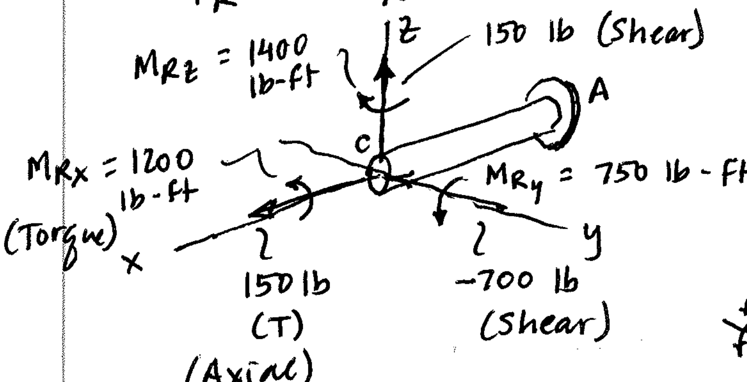
Forces shown in Scalar Form:



Worked as a Resultant Problem (my preference)

Replace this system of forces ($\vec{F}_1$ and $\vec{F}_2$) as an $\vec{F}_R + \vec{M}_R$ at the cut C. Move $\vec{F}_R$ to C.

$\vec{F}_R = \vec{F}_1 + \vec{F}_2 = [350, -700, 150]$ lb

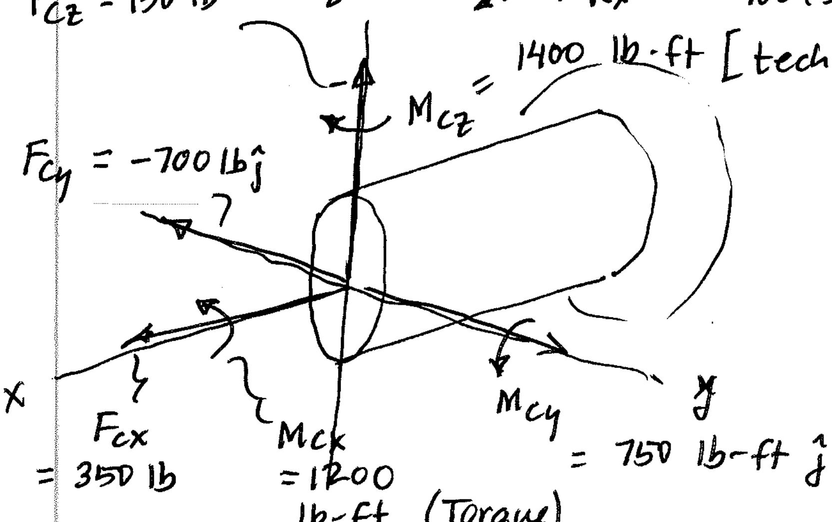


Preserve the bending moments of the forces from their previous locations.

$M_{R_y} = 350(3) - 150(2)$
$= 1050 - 300 = 750$ lb-ft

$M_{R_z} = 400(2) + 300(2) = 1400$ lb-ft

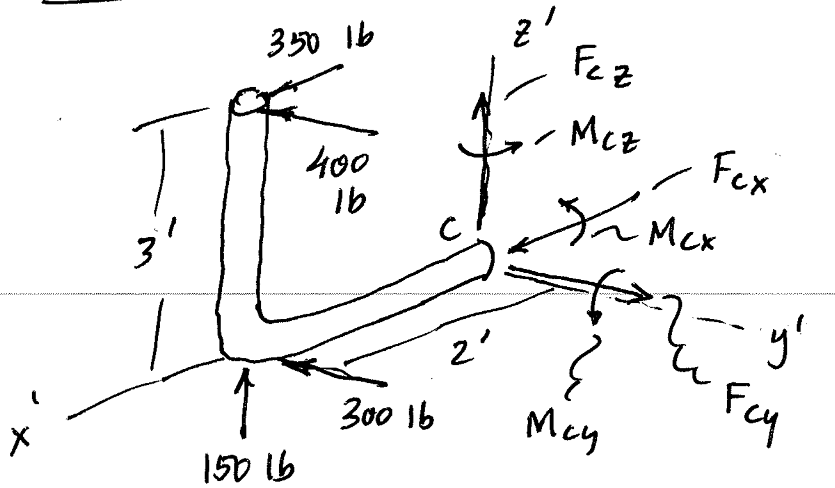
$M_{R_x} = 400(3) = 1200$ lb-ft



} Larger Picture..

7-41 (cont'd)

Worked as an Equilibrium Problem



Steps:

① Draw FBD.

② Draw $\vec{F}_c$ and $\vec{M}_c$ comps in their + sense

③ Write Equil Eqns:

$\swarrow^{+} \Sigma F_x = 0;$ $F_{cx} + 350 = 0;$ $\boxed{F_{cx} = -350 \text{ lb}}$

$= 350$ lb (Tension) (Axial)

$\searrow^{+} \Sigma F_y = 0;$ $F_{cy} - 400 - 300 = 0;$ $\boxed{F_{cy} = +700 \text{ lb}}$

This is a shear force.

$+\uparrow \Sigma F_z = 0;$ $F_{cy} + 150 = 0;$ $\boxed{F_{cy} = -150 \text{ lb}}$

$\overset{+}{\swarrow} \Sigma M_x = 0;$ $400(3) + M_{cx} = 0;$ $\boxed{M_{cx} = -1200 \text{ lb·ft}}$

This is a torque.

$\overset{+}{\searrow} \Sigma M_y = 0;$ $M_{cy} + 350(3) - 150(2) = 0;$

$\boxed{M_{cy} = -750 \text{ lb-ft}}$

This is a bending moment.

$+\uparrow \Sigma M_z = 0;$ $M_{cz} - 400(2) - 300(2) = 0;$

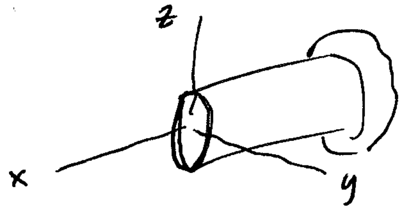
$\boxed{M_{cz} = +1400 \text{ lb-ft}}$

All the same answers, but negative (equal + opposite).

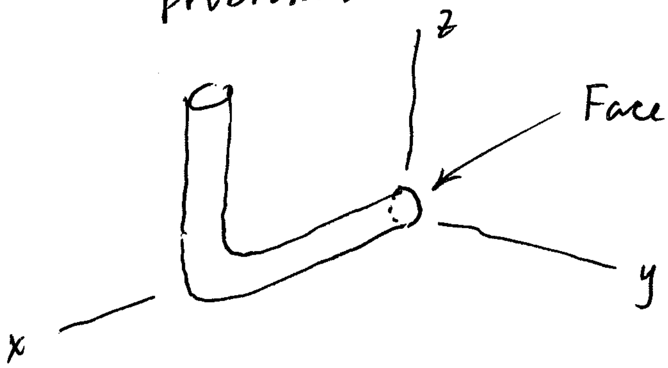
7-41 (cont'd)

To me, the advantages of working as a "Resultant" problem:

1. Just more forces.
2. compute their resultant moments
3. Forces + moments easy to visualize on a face you can **see**.



Disadvantages of working as an equilibrium problem:



Face you cannot see!

Stresses you cannot see.

Have to write the 6 $\Sigma F = 0$, $\Sigma M = 0$ eqns.